

MARITIME CRIME INVESTIGATION & FORENSIC ATTRIBUTION REPORT

COAST GUARD REGIONAL HEADQUARTERS (WEST) / NATIONAL TECHNICAL RESEARCH ORGANISATION (NTRO)

1. INCIDENT OVERVIEW & SATELLITE DETECTION TELEMETRY

Incident Identifier (Join Key):	SPELL-2026-ARABIAN-001	Surveillance Satellite:	S1C_IW_GRDH_00000
Detection Timestamp (UTC):	2026-08-25T06:00:00Z	SAR Slick Area:	1.44 km ²
Slick Centroid (WGS84):	18.7999°N, 72.4953°E	Thickness Classification:	SHEEN
Jurisdiction Zone:	Indian EEZ (200 NM Sovereign Shelf)	Look-Alike Suppressed:	CONFIRMED (Mineral Oil Signature)

2. LAGRANGIAN OCEAN DRIFT HINDCAST & PROBABILITY ENVELOPES

Advection hindcast executed using OpenDrift 1.14.11 (OpenOil module) with Runge-Kutta 4th Order (RK4) numerical integration. Forcing fields: Copernicus Marine GLORYS 0.083° surface currents and ECMWF ERA5 10m hourly wind vectors. Ensemble configuration: 25 independent physical members with NOAA GNOME windage distributions ($\alpha \sim N(0.030, 0.004)$) and horizontal turbulent diffusivity (10 m²/s). Origin probability envelopes calculated via 2D Gaussian Kernel Density Estimation (KDE).

3. RANKED SUSPECT VESSEL ADJUDICATION ROSTER

Rank	Vessel Name / MMSI	IMO / Flag	Voyage Ports	AIS Blackout	Score
#1	TRIDENT II MMSI: 419001415	IMO: 9412345 Flag: LBR	Fujairah (UAE) → Vadinar (IND)	None	0.8842
#2	PACIFIC VOYAGER MMSI: 419702000	IMO: 9382104 Flag: IND	Jebel Ali (UAE) → Mumbai (IND)	None	0.4820
#3	MV MAERSK KARACHI MMSI: 419900753	IMO: 9194276 Flag: DNK	JNPT Mumbai (IND) → Salalah (OMN)	None	0.1620
#4	MOPU SAGAR SAMRAT MMSI: 419381000	IMO: 8751485 Flag: IND	Bombay High Field → Stationary	None	0.1210
#5	SEA CHALLENGER MMSI: 636022539	IMO: 9625345 Flag: LBR	Kandla (IND) → Singapore (SGP)	None	0.1840
#6	VIRTUE-1 MMSI: 563013000	IMO: 9421045 Flag: SGP	Ras Laffan (QAT) → Mumbai (IND)	None	0.2940

4. PRIMARY SUSPECT FORENSIC AUDIT: TRIDENT II (419001415)

PRIMARY SUSPECT: TRIDENT II (Crude Oil Tanker, MMSI 419001415) transiting Fujairah to Vadinar severed its AIS transmission for 4.8 hours across local nighttime while passing directly through the 90% drift origin cone at 2026-08-23T20:15 UTC. Vessel decelerated to 6.2 knots during blackout (characteristic bilge discharge profile). 4D reconstructed route achieves 89.1% spatial intersection, and forward confession advection matches the observed SAR slick with 74.2% IoU.

Proximity to Origin Cone:	92.40%	Route Ray-Trace Match:	89.12%
Confession Advection IoU:	74.20%	Dominant Anomaly Factor:	discharge_speed_fraction

5. STATUTORY ADMIRALTY AUTHORITY & CRYPTOGRAPHIC CHAIN-OF-CUSTODY

Statutory Jurisdiction: Republic of India Territorial Waters, Continental Shelf, Exclusive Economic Zone and Other Maritime Zones Act 1976 (Act No. 80 of 1976), Section 7. Enforcement authorized pursuant to United Nations Convention on the Law of the Sea (UNCLOS) Article 211(5) and MARPOL Annex I.

Advisory Directive: INDIAN COAST GUARD MARITIME RESCUE COORDINATION CENTRE (MRCC) INTERCEPT DISPATCH RECOMMENDED.

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